Q1 JEE Main 2020 - 2 September (Morning)

The mass density of a spherical galaxy varies as $\frac{\mathrm{K}}{\mathrm{r}}$ over a large distance r ' from its centre.
In that region, a small star is in a circular orbit of radius R. Then the period of revolution, T depends on R as

(A) $T^2 \propto \frac{1}{R^3}$
(B) $\mathrm{T}^2 \propto \mathrm{R}$
(C) $\mathrm{T} \propto \mathrm{R}$
(D) $\mathrm{T}^2 \propto \mathrm{R}^3$

Q2 JEE Main 2020 - 2 September (Evening)

The height 'h' at which the weight of a body will be the same as that at the same depth 'h' from the surface of the earth is (Radius of the earth is R and effect of the rotation of the earth is neglected)

(A) $\frac{\sqrt{5}\mathrm{R}-\mathrm{R}}{2}$
(B) $\frac{\sqrt{3}\mathrm{R}-\mathrm{R}}{2}$
(C) $\frac{R}{2}$
(D) $\frac{\sqrt{5}}{2}\mathrm{R}-\mathrm{R}$

Q3 JEE Main 2020 - 3 September (Morning)

A satellite is moving in a low nearly circular orbit around the earth. Its radius is roughly equal to that of the earth's radius $\mathrm{R_e}$. By firing rockets attached to it, its speed is instantaneously increased in the direction of its motion so that it become $\sqrt{\frac{3}{2}}$ times larger. Due to this the farthest distance from the centre of the earth that the satellite reaches is R. Value of R is

(A) $3\mathrm{R_e}$
(B) $4R_e$
(C) $2.5\mathrm{R_e}$
(D) $2\mathrm{R_e}$

Q4 JEE Main 2020 - 3 September (Evening)

The mass density of a planet of radius R varies with the distance r from its centre as $\rho(\mathbf{r})=\rho_0\left(1-\frac{\mathbf{r}^2}{\mathbf{R}^2}\right)$ · Then the gravitational field is maximum at

(A) $\mathbf{r}=\frac{1}{\sqrt{3}}\mathbf{R}$
(B) $\mathrm{r}=\sqrt{\frac{3}{4}}\mathrm{R}$
(C) $\mathbf{r}=\sqrt{\frac{5}{9}}\mathbf{R}$
(D) $r=R$

Q5 JEE Main 2020 - 4 September (Morning)

On the x -axis and at a distance x from the origin, the gravitational field due to a mass distribution is given by $\frac{Ax}{\left(x^2+a^2\right)^{3/2}}$ in the x-direction. The magnitude of gravitational potential on the x-axis at a distance x, taking its value to be zero at infinity, is

(A) $A\left(x^2+a^2\right)^{3/2}$
(B) $\frac{A}{\left(x^2+a^2\right)^{3/2}}$
(C) $\frac{A}{\left(x^2+a^2\right)^{1/2}}$
(D) $\mathrm{A}\left(\mathrm{x}^2+\mathrm{a}^2\right)^{1/2}$

Q6 JEE Main 2020 - 4 September (Evening)

A body is moving in a low circular orbit about a planet of mass M and radius R. The radius of the orbit can be taken to be R itself. Then the ratio of the speed of this body in the orbit to the escape velocity from the planet is

(A) 1
(B) $\sqrt{2}$
(C) 2
(D) $\frac{1}{\sqrt{2}}$

Q7 JEE Main 2020 - 5 September (Morning)

The value of the acceleration due to gravity is g_1 at a height $h=\frac{R}{2}$ ($R=$ radius of the earth) from the surface of the earth. It is again equal to g_1 at a depth d below the surface of the earth. The ratio $\left(\frac{d}{R}\right)$ equals

(A) $\frac{5}{9}$
(B) $\frac{1}{3}$
(C) $\frac{7}{9}$
(D) $\frac{4}{9}$

Q8 JEE Main 2020 - 5 September (Evening)

The acceleration due to gravity on the earth's surface at the poles is g and angular velocity of the earth about the axis passing through the pole is ω. An object is weighed at the equator and at a height h above the poles by using a spring balance. If the weights are found to be same, then h is $(h << R$, where R is the radius of the earth)

(A) $\frac{R^2\omega^2}{2g}$
(B) $\frac{R^2\omega^2}{g}$
(C) $\frac{R^2\omega^2}{8g}$
(D) $\frac{R^2\omega^2}{4g}$

Q9 JEE Main 2020 - 6 September (Morning)

A satellite is in an elliptical orbit around a planet P. It is observed that the velocity of the satellite when it is farthest from the planet is 6 times less than that when it is closest to the planet. The ratio of distances between the satellite and the planet at closest and farthest points is

(A) 01:06
(B) 01:02
(C) 03:04
(D) 01:03

Q10 JEE Main 2020 - 6 September (Evening)

Two planets have masses M and $16\mathrm{M}$ and their radii are a and $2\mathrm{a}$, respectively. The separation between the centres of the planets is 10 a. A body of mass m is fired from the surface of the larger planet towards the smaller planet along the line joining their centres. For the body to be able to reach at the surface of smaller planet, the minimum firing speed needed is

(A) $\frac{3}{2}\sqrt{\frac{5\mathrm{GM}}{\mathrm{a}}}$
(B) $\sqrt{\frac{\mathrm{GM}^2}{\mathrm{ma}}}$
(C) $2\sqrt{\frac{\mathrm{GM}}{\mathrm{a}}}$
(D) $4\sqrt{\frac{\mathrm{GM}}{\mathrm{a}}}$

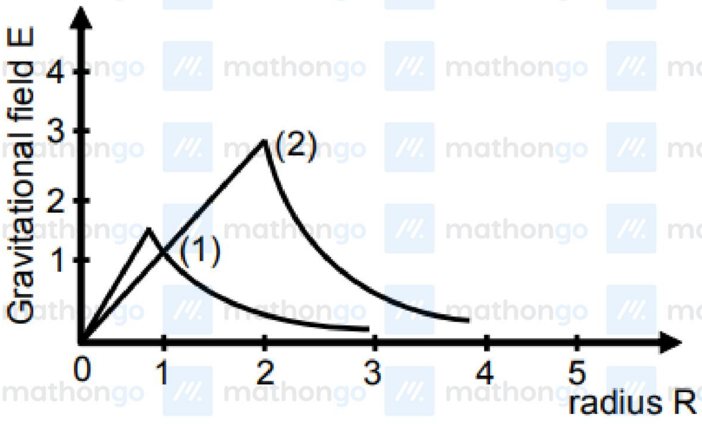
Q11 JEE Main 2020 - 7 January (Evening)

A box weighs $196\,\mathrm{N}$ on a spring balance at the north pole. Its weight recorded on the same balance if it is shifted to the equator is close to (Take $g = 10ms^{-2}$ at the north pole and the radius of the earth = $6400\,km$)

(A) $194.32\ N$
(B) $194.66\ N$
(C) $195.32\ N$
(D) $195.66\ N$

Q12 JEE Main 2020 - 8 January (Morning)

Two spherical bodies of mass m_1 and m_2 have radii $1m$ and $2m$ respectively. The gravitational field of the two bodies with the radial distance from centre is shown below. The value of $\frac{m_1}{m_2}$ is



(A) $\frac{1}{6}$
(B) $\frac{1}{3}$
(C) $\frac{1}{2}$
(D) $\frac{1}{4}$

Q13 JEE Main 2020 - 8 January (Evening)

An asteroid is moving directly towards the centre of the earth. when at a distance of $10R$ (R is the radius of the earth) from the earths centre, it has a speed of $12km/s$. Neglecting the effect of earths atmosphere, what will be the speed of the asteroid when it hits the surface of the earth (escape velocity from the earth is $11.2\,km/s^{-1}$)? Give your answer to the nearest integer in kilometer/s

Q14 JEE Main 2020 - 9 January (Morning)

A body A of mass m is moving in a circular orbit of radius R about a planet. Another body B of mass $\frac{m}{2}$ collides with A with a velocity which is half $\left(\frac{\vec{v}}{2}\right)$ the instantaneous velocity $\vec{v}$ of A. The collision is completely inelastic. Then, the combined body :

(A) Circular
(B) Elliptical
(C) Straight line
(D) Fall directly below on the ground

Q15 JEE Main 2020 - 9 January (Evening)

Planet A has mass M and radius R. Planet B has half the mass and half the radius of Planet A. If the escape velocities from the Planets A and B are V_A and V_B, respectively, then $v_A/v_B = n/4$. The value of n is :

(A) 3
(B) 2
(C) 1
(D) 4

Answer Key

Q1 (B)	**Q2** (A)	**Q3** (A)	**Q4** (C)
Q5 (C)	**Q6** (D)	**Q7** (A)	**Q8** (A)
Q9 (A)	**Q10** (A)	**Q11** (C)	**Q12** (A)
Q13 (16)	**Q14** (B)	**Q15** (D)	

Q1

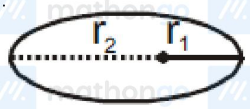
$m\omega^2 R = \frac{GMm}{R^2}$
$M = \int_0^R 4\pi r^2 dr \left(\frac{K}{r}\right) = 2\pi K R^2$
$\omega^2 R = 2G\pi K$
$T^2 \propto R$

Q2

$\frac{GM}{(h+R)^2} = \frac{GM}{R^2}\left(1 - \frac{h}{R}\right)$
$\mathbf{R}^3 = (\mathbf{h} + \mathbf{R})^2(\mathbf{R} - \mathbf{h})$
Solving $h = \frac{\sqrt{5}-1}{2}R$

Q3

Total energy $E = \frac{-GMm}{2a}$
$\therefore \quad \frac{1}{2}m \times \left(\frac{2}{3}\frac{GM}{R}\right) - \frac{GMm}{R} = \frac{-GMm}{2a}$
$\Rightarrow \quad a = 2R$
Major axis $= 4R$



$r_1 = R_1 \Rightarrow r_2 = 3R$

Q4

$\rho(\mathbf{r}) = \rho_0\left(1 - \frac{\mathbf{r}^2}{\mathbf{R}^2}\right)$
$\therefore \quad m = \int_0^r \rho(r) \times 4\pi r^2 dr$
$= 4\pi\rho_0\left(\frac{r^3}{3} - \frac{r^5}{5R^2}\right)$

$$\therefore \quad \mathrm{E}=\frac{\mathrm{Gm}}{\mathrm{r}^2}=4\pi\rho_0\left[\frac{\mathrm{r}}{3}-\frac{\mathrm{r}^3}{5\mathrm{R}^2}\right]$$

$$\therefore \quad \frac{\mathrm{dE}}{\mathrm{dr}}=0 \Rightarrow \mathrm{r}=\sqrt{\frac{5}{9}}\mathrm{R}$$

Q5

$$\int_0^v dV=-\int_{-\infty}^x E \cdot dx$$

$$V=-A\int_{-\infty}^x \frac{x dx}{\left(x^2+a^2\right)^{3/2}}$$

$$\mathrm{V}=\frac{\mathrm{A}}{\left(\mathrm{x}^2+\mathrm{a}^2\right)^{1/2}}$$

Q6

$$v_0=\sqrt{\frac{GM}{R}}$$

$$\mathrm{v_e}=\sqrt{\frac{2\mathrm{GM}}{\mathrm{R}}}$$

$$\frac{u_0}{u_e}=\frac{1}{\sqrt{2}}$$

Q7

$$g\left(h=\frac{R}{2}\right)=\frac{GM}{\left(\frac{3R}{2}\right)^2}=\frac{4g}{9}$$

$$g\left(1-\frac{d}{R}\right)=\frac{4g}{9}$$

$$\mathrm{d}=\frac{5\mathrm{R}}{9}$$

Q8

$$\mathrm{g_{equator}}=\mathrm{g}-\omega^2\mathrm{R}$$

$$\mathrm{g_h}=\mathrm{g}\left(1-\frac{2\mathrm{h}}{\mathrm{R}}\right)$$

$$g\left(\frac{2h}{R}\right)=\omega^2 R$$

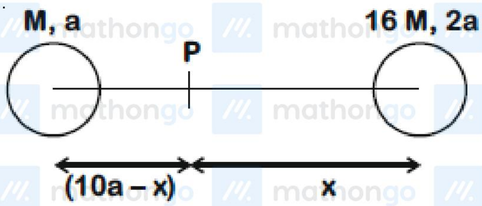
$$\mathrm{h}=\frac{\omega^2\mathrm{R}^2}{2\mathrm{g}}$$

Q9

$\frac{V_{\max}}{V_{\min}} = \frac{(1+e)}{(1-e)}$

$\frac{r_{\max}}{r_{\min}} = \frac{(1+e)}{1-e} = 6$

Q10



Velocity should be given so as to reach a point where field is zero.

$\frac{16\mathrm{M}}{\mathrm{x}^2} = \frac{\mathrm{M}}{(10\mathrm{a}-\mathrm{x})^2}$

$x = 8a$

By CoE

$-\frac{GMm}{8a} - \frac{16GMm}{2a} + \frac{1}{2}mv^2 = -\frac{16GMm}{8a} - \frac{GMm}{2a}$

$v = \frac{3}{2}\sqrt{\frac{5\mathrm{Gm}}{a}}$

Q11

$\text{at pole, weight} = mg = 196$

$m = 19.6\ kg$

$\text{at equator, weight} = mg - m\omega^2 R$

$= 196 - (19.6)\left[\frac{2\pi}{24 \times 3600}\right]^2 \times 6400 \times 10^3$

$= 195.33\ N$

Q12

$$3 = \frac{Gm_2}{2^2}$$
$$2 = \frac{Gm_1}{1^2}$$
$$\therefore \quad \frac{3}{2} = \frac{1}{4}\frac{m_2}{m_1}$$
$$\frac{m_1}{m_2} = \frac{1}{6}$$

Q13

$$KE_i + PE_i = KE_f + PE_f$$
$$\frac{1}{2}m{u_0}^2 + \left(-\frac{GMm}{10R}\right) = \frac{1}{2}mv^2 + \left(-\frac{GMm}{R}\right)$$
$$v = \sqrt{u_0^2 + \frac{9}{5}\frac{GM}{R}}$$
$$= \sqrt{12^2 + \left(\frac{9}{5}\right)\frac{(11.2)^2}{2}}$$
$$= \sqrt{144 + 0.9(11.2)^2} = \sqrt{256.896}$$
$$= 16.028\ km/s$$
$$\simeq 16$$

Q14

Conserving momentum

$$\frac{m}{2}\frac{v}{2} + mv = \left(m + \frac{m}{2}\right)v_f$$
$$v_f = \frac{5\ mV}{4 \times \frac{3m}{2}} = \frac{5V}{6}$$

$v_f = v_{orb}\ (= v)$ thus the combined mass will go on to an elliptical path

Q15

$$v_e = \sqrt{\frac{2GM}{R}}$$

$$\therefore \quad \frac{v_1}{v_2} = \frac{\sqrt{\frac{2GM}{R}}}{\sqrt{\frac{2GM/2}{R/2}}} = 1 = \frac{n}{4}$$

$$\Rightarrow \quad n = 4$$